Method	Low-Level				High-Level				Brain Correlation		
	PixCorr	SSIM	Alex(2)	Alex(5)	Incep	CLIP	Eff	SwAV	Early Vis.	Higher Vis.	Visual Cortex
Mental Imagery Reconstructions											
MIRAGE (ours)	–	–	–	–	–	–	–	–	–	–	–
MindEye1	0.105	**0.000**	**0.017**	0.414	0.917	**0.049**	**0.000**	0.060	**0.001**	0.345	**0.035**
Brain Diffuser	**0.000**	0.781	**0.000**	**0.023**	0.826	**0.037**	**0.000**	0.092	**0.000**	**0.038**	**0.000**
iCNN	0.675	**0.000**	**0.000**	**0.000**	0.173	**0.000**	**0.000**	0.005	**0.000**	**0.000**	**0.000**
MindEye2	**0.000**	0.202	**0.000**	**0.000**	**0.005**	**0.001**	**0.000**	**0.006**	**0.000**	**0.000**	**0.000**
Vision Reconstructions											
MIRAGE (ours)	–	–	–	–	–	–	–	–	–	–	–
MindEye1	0.816	**0.014**	**0.000**	0.001	**0.000**	0.463	0.845	0.000	0.295	0.365	0.585
Brain Diffuser	**0.000**	0.331	**0.000**	**0.013**	**0.000**	**0.000**	**0.000**	0.115	**0.000**	**0.002**	**0.000**
iCNN	0.748	**0.000**	**0.000**	0.000	**0.000**	**0.000**	**0.000**	0.000	0.000	0.109	0.007
MindEye2	**0.000**	0.002	**0.000**	0.490	**0.000**	0.670	0.361	0.000	0.310	**0.017**	**0.005**

Table 1. P-values from t-tests comparing image feature metrics of MIRAGE (ours) with alternative methods. Bolded p-values indicate that MIRAGE significantly outperforms ($p < 0.05$ and $t > 0$) for metrics where higher is better, ($p < 0.05$ and $t < 0$) for metrics where lower is better.

Method	All Stimuli ↑	T	P
Mental Imagery Reconstructions			
MIRAGE (ours)	**78.30%**	-	-
MindEye	73.00%	2.327	**0.020**
Brain Diffuser	<u>73.95%</u>	1.983	**0.048**
iCNN	66.15%	5.139	**0.000**
MindEye2	56.96%	8.784	**0.000**
Vision Reconstructions			
MIRAGE (ours)	<u>84.22%</u>	-	-
MindEye	**84.29%**	-0.028	0.978
Brain Diffuser	80.13%	1.638	0.102
iCNN	74.52%	3.698	**0.000**
MindEye2	83.05%	0.485	0.628

Table 2. T-test results on our human identification accuracy experiment. The "T" column reports the t-test statistic from t-tests comparing MIRAGE to each alternative method. The "P" column reports the corresponding p-value. Bolded p-values indicate that MIRAGE significantly outperforms the alternative ($p < 0.05$ and $t > 0$).